Open book exam. Submit your answers to gradescope.
In your explanations and justifications, write complete and grammatically correct sentences.
Please do not ask questions during the exam.

3. Consider the sequence $S = [a, b, c, d, b, b, e, d, c, c]$ for processing using a cache of size 2.

   (a) Show all steps in the farthest-in-the-future algorithm on S.

   (b) Use this example to illustrate that a nonreduced schedule does not lead to a number of cache misses fewer than the farthest-in-the-future eviction schedule.

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Answer:

3. (a) We run the farthest-in-the-future algorithm:

      i. a causes miss in cache $[\ ,\ ]$, insert a, new cache is $[a,\ ]$
      ii. b causes miss in cache $[a,\ ]$, insert b, new cache is $[a, b]$
      iii. c causes miss in cache $[a, b]$, evict a, new cache is $[c, b]$
      iv. d causes miss in cache $[c, b]$, evict c, new cache is $[d, b]$
      v. b is already in cache $[d, b]$
      vi. b is already in cache $[d, b]$
      vii. e causes miss in cache $[d, b]$, evict b, new cache is $[d, e]$
      viii. d is already in cache $[d, e]$
      ix. c cause cache miss in $[d, e]$, evict d, new cache $[c, e]$
      x. c is already in cache $[c, e]$

   (b) As an example of a nonreduced schedule, consider the replacement of d by e in step 5 or 6, before it is needed in step 7.

      In that case, we would not have had a cache miss in step 7, but then we would have had a cache miss in step 8 when d was needed.

      In this example of a nonreduced schedule, the total number of cache missed remains the same as the fartest-in-the-future eviction schedule.